Type MS · Match digestive enzymes to substrate, product and site

Answer 1 of 12 · Question 2 part a.i · 3 marks

2023 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

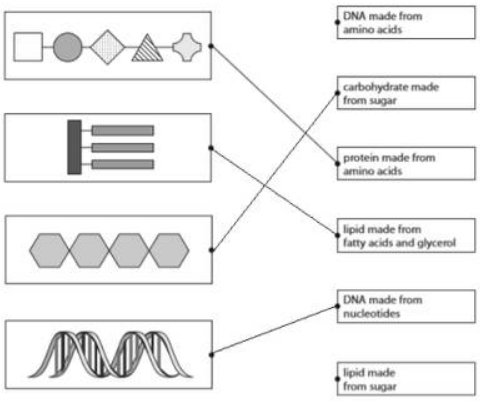
2 (a)	(i)	in the following order: • proteases; • protein Q; • amino acids;	Reject protein P	3
	(ii)	stomach/ small intestine/ pancreas/ duodenum/ ileum;		1
	(iii)	DNA / mRNA / messenger RNA;		1
(b)	(i)	• as pH increases activity increases; • to pH2/peaks at pH2/to optimum; • decreases from pH2;	Allow increases and then decreases for one mark	3 exp
	(ii)	Any two from: • enzyme is denatured (at pH2); • change in shape of active sites; • no enzyme-substrate complexes formed/no binding with substrate;		2 exp
	(iii)	• repeat; • calculate a mean/average/remove anomalous result		2 grad
	(iv)	Any one from: • temperature; • substrate concentration/volume; • enzyme concentration/volume;		1 grad

Total for Question 2 = 13 Marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 2 of 12 · Question 3 part b · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

3	(a)		Do not allow more than one line from each structure	3
	(b)	**D** stomach;		1
	(c) (i)	3/three		1
	(ii)	not washing equipment/transfer of food containing protein into distilled water;	Allow any valid alternative	1
	(d)	Broken glass - cuts: Clear up breakages using a dustpan and brush/keep equipment away from edge of tables;	Ignore wear gloves	1
		Biuret reagent - irritant: Wear goggles/wash hands after use/don't use if skin is sensitive;	Ignore wear gloves	1

Total question 3 = 8 marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 2 of 12 · Question 3 part b · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 3 of 12 · Question 1 part b.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

Question number				
1	(a)	any three from • oesophagus (1) • stomach (1) • small intestine/ileum/duodenum (1) • large intestine/rectum (1)		3
	(b) (i)	protease/pepsin/trypsin (1)		1
	(ii)	• chewing/grinding/mastication (1) • (mix with) saliva (1) • digestion of starch (1) • to maltose (1)		4
	(c)	any three from • provides bulk (1) • helps peristalsis (1) • prevents constipation (1) • reduces risk of bowel cancer (1)		3

Total for question = 11 marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 3 of 12 · Question 1 part b.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 4 of 12 · Question 2 part a.i · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

2 (a) (i)	C; (lipase)		1
(ii)	A; (carbon, hydrogen, oxygen)		1
(iii)	C; (liver)		1
(b)	• show effect of enzyme (1) • show that bile salts on their own don't break down fat (1)	Ignore control	2
(c)	• create alkaline/optimum pH (1) • for enzyme to work quickly (1)	Allow increases/adjusts pH	2
(d)	• increase rate of digestion (1) • with no bile salts change occurred after 15 minutes/digested more slowly in tube A (1) • occurred after 5 minutes with bile salts present/digested more quickly in tube B (1)		3

Total 10 marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 4 of 12 · Question 2 part a.i · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 5 of 12 · Question 2 part b.i · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
2	(a) (i)	W large intestine; X oesophagus; Y stomach; Z small intestine;	Allow colon Allow ileum	4
	(ii)	In the following order only: • bolus; • muscles; • contract; • peristalsis;		4
	(b) (i)	salivary gland/ pancreas;		1
	(ii)	(lining of) **small** intestine/ ileum/ duodenum;		1
	(iii)	• starch digested/broken down to sugar/glucose; • (sugar) diffuses/moves out through (Visking tubing);	Allow maltose	2
	(iv)	Benedict's solution/reagent;		1
	(v)	Any two from the following: • blue/black colour; • (undigested) starch is present; • too large to pass through Visking tubing;		2

Total marks 15

Type MS · Match digestive enzymes to substrate, product and site

Answer 5 of 12 · Question 2 part b.i · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 6 of 12 · Question 4 part a.ii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number				
4	(a) (i)	Any three of the following: • add alcohol/ethanol to food; • shake; • add water; • milky coloration shows fat present;	Allow rub food; onto filter paper; translucent; answer must be in sequence i.e. ethanol first	3
	(ii)	lipase		1
	(iii)	fatty acids; glycerol;		2
	(b) (i)	Any value between 8.6 and 8.8;		1
	(ii)	independent = time/ temperature; dependent = pH;		2
	(iii)	• 37°C is the optimum temperature; • lipase/enzyme is more active/works faster; • fat is digested more quickly; • at 80 °C/higher temperature enzyme becomes denatured;	Allow reverse argument throughout: • at 80°C/higher temperature enzyme is denatured; • enzyme less active; • fat digested more slowly; • 37°C is the optimum temperature (for enzyme activity/enzyme works faster at 37°C	4

Total marks 13

Type MS · Match digestive enzymes to substrate, product and site

Answer 6 of 12 · Question 4 part a.ii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 7 of 12 · Question 4 part a.iii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

4 (a) (i)	Any three of the following: • add alcohol/ethanol to food; • shake; • add water; • milky coloration shows fat present;	Allow rub food; onto filter paper; translucent; answer must be in sequence i.e. ethanol first	3
(ii)	lipase		1
(iii)	fatty acids; glycerol;		2
(b) (i)	Any value between 8.6 and 8.8;		1
(ii)	independent = time/ temperature; dependent = pH;		2
(iii)	• 37°C is the optimum temperature; • lipase/enzyme is more active/works faster; • fat is digested more quickly; • at 80 °C/higher temperature enzyme becomes denatured;	Allow reverse argument throughout: • at 80°C/higher temperature enzyme is denatured; • enzyme less active; • fat digested more slowly; • 37°C is the optimum temperature (for enzyme activity/enzyme works faster at 37°C	4

Total marks 13

Type MS · Match digestive enzymes to substrate, product and site

Answer 7 of 12 · Question 4 part a.iii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 8 of 12 · Question 2 part a.ii · 1 mark

2023 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

2 (a)	(i)	in the following order: • proteases; • protein Q; • amino acids;	Reject protein P	3
	(ii)	stomach/ small intestine/ pancreas/ duodenum/ ileum;		1
	(iii)	DNA / mRNA / messenger RNA;		1
(b)	(i)	• as pH increases activity increases; • to pH2/peaks at pH2/to optimum; • decreases from pH2;	Allow increases and then decreases for one mark	3 exp
	(ii)	Any two from: • enzyme is denatured (at pH2); • change in shape of active sites; • no enzyme-substrate complexes formed/no binding with substrate;		2 exp
	(iii)	• repeat; • calculate a mean/average/remove anomalous result		2 grad
	(iv)	Any one from: • temperature; • substrate concentration/volume; • enzyme concentration/volume;		1 grad

Total for Question 2 = 13 Marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 9 of 12 · Question 4 part a.vi · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.vi in this mark scheme extract

Question number				
4 (a)	(i)	• axes labels and units and right way around; • linear scale and at least half grid; • correctly plotted points to half square tolerance;		(3)
	(ii)	• curve of best fit;	do not accept straight line going through all points	(1)
	(iii)	6		(1)
	(iv)	• time taken to digest starch decreases until pH 6; • time taken to digest starch increases (from/after PH6) to pH10;	Allow 1 mark for decrease and then increase in time as pH increases	(2)
	(v)	• reference to pH2 not being optimum/best pH for **enzyme** (activity)/pH6 being (closer to) optimum pH for enzyme; • enzymes denatured / active site changes shape/deformed (at pH2); • less/no binding to substrate/ substrate cannot fit into active site/ fewer enzyme -substrate complex;		(3)
	(vi)	small intestine/ mouth/duodenum;		(1)
(b)		• add iodine solution; • blue-black/blue/black colour indicates the presence of starch;		(2)

Total for Question 4 = 13 marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 10 of 12 · Question 1 part b · 4 marks

2024 JUN · Paper 02 (regional R) · find part b in this mark scheme extract

1 (a) (i)		X = incisor(s); Z = molar(s)/pre-molar(s);		(2)
	(ii)	tearing/ripping/gripping/holding (food);		(1)
(b)		• amylase/carbohydrase; • sugar/maltose; • any value between 6.7 and 7.4; • decrease/reduce/stop/slow down/take more time;	allow glucose	(4)
(c) (i)		oesophagus;		(1)
	(ii)	• muscular contractions/peristalsis; • push food/bolus (downwards/towards stomach);		(2)

Total for Question 1 = 10 marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 11 of 12 · Question 7 part f · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part f in this mark scheme extract

7 (a)	• body temperature; • optimum/enzyme is not denatured;	Accept alimentary for MP1	2
(b)	• effect of pH/to see if pH affects enzyme activity; • range of pH used/different pH used/ pH is the independent variable;	***Effect of different pH on enzyme activity = 2 marks***	2
(c)	Any four from: Answers about validity(MP1-4) • The same temperature; • The same volume of enzyme/suspension; • not (valid) because different time used; • the same enzyme is used; Answers about accuracy(MP 5-7) • as difficult to judge endpoints/ when it goes clear; • students' judgement of endpoint may vary/ subjective/ not standardised/ did not use a colorimeter; • timing is too short to reach endpoint/ cannot tell it will go clear if left for longer;	They do not have to mention validity/accuracy to score marks. Must have at least one mark from each section to score 4. Maximum of 3 if answers are from one section	4
(d)	• tube with no enzyme/ denatured enzyme/boiled enzyme; • same volume of boiled enzyme/water /5cm^3 of water/boiled enzyme; • same volume of suspension/same overall volume/ same volume of substrate (in test tube);		3
(e)	• optimum pH; • fastest between 2 and 4;		2
(f)	• stomach; • acidic/low pH/pH2-4/pH2;		2

Total 15 marks

Type MS · Match digestive enzymes to substrate, product and site

Answer 11 of 12 · Question 7 part f · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part f in this mark scheme extract

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Type MS · Match digestive enzymes to substrate, product and site

Answer 12 of 12 · Question 7 part a · 1 mark

2025 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

7	(a)	amylase;		1
	(b)	Any two from • enzymes are immobilised on testing strip; • react with/breaks down glucose (in urine); • (catalyse a chemical reaction that) causes a colour change (in strip);		2
	(c)	pancreas;		1
	(d)	• rejection/immune response/allergies; • (immune system) recognised it as foreign/antigen;		2
	(e)	any two from • use restriction enzymes; • cut out insulin gene from human DNA; • insert human insulin gene into yeast genome/plasmid/genetic material;		2
			Total for Question 7 = 8 marks	